# QuickPhone — Logical ERD (21 entities)

Crow's foot notation. Header colour and tag = the DFD data store that holds the entity (see the store–entity matrix).

### ROLE
D10 Staff & Stores

| | | |
|---|---|---|
| PK | **RoleID** | int |
| UK | RoleName | string |
| | Description | string |

### STORE
D10 Staff & Stores

| | | |
|---|---|---|
| PK | **StoreID** | int |
| UK | StoreName | string |
| | Address | string |
| | Phone | string |
| | EmailAddress | string |
| | BusinessHours | string |

### EMPLOYEE
D10 Staff & Stores

| | | |
|---|---|---|
| PK | **EmployeeID** | int |
| FK *store* | StoreID | int |
| FK *role* | RoleID | int |
| | FirstName | string |
| | LastName | string |
| UK | Email | string |
| | ActiveFlag | boolean |

### CUSTOMER
D1 Customers & Devices

| | | |
|---|---|---|
| PK | **CustomerID** | int |
| | FirstName | string |
| | LastName | string |
| IX | Phone | string |
| | Email | string |
| | CreatedAt | datetime |

### DEVICE
D1 Customers & Devices

| | | |
|---|---|---|
| PK | **DeviceID** | int |
| FK *customer* | CustomerID | int |
| | DeviceType | string |
| | Brand | string |
| | Model | string |
| IX | SerialOrIMEI | string |

### APPOINTMENT
D2 Appointments

| | | |
|---|---|---|
| PK | **AppointmentID** | int |
| FK *customer* | CustomerID | int |
| FK *store* | StoreID | int |
| FK *employee* | TechnicianID | int |
| | ScheduledAt | datetime |
| | ApptStatus | string |

### JOB_STATUS
D3 Repair Jobs

| | | |
|---|---|---|
| PK | **StatusID** | int |
| UK | StatusName | string |
| | DisplayOrder | int |

### REPAIR_JOB
D3 Repair Jobs

| | | |
|---|---|---|
| PK | **RepairJobID** | int |
| UK | TicketNumber | string |
| FK *device* | DeviceID | int |
| FK *store* | StoreID | int |
| FK,UK *appt.* | AppointmentID | int |
| FK *employee* | IntakeEmployeeID | int |
| FK *employee* | AssignedTechID | int |
| FK *job_status* | CurrentStatusID | int |
| | ReportedIssue | string |
| | IntakeCondition | string |
| | CheckInAt | datetime |
| | FinalCost | decimal |
| | CompletedAt | datetime |
| | PickedUpAt | datetime |
| FK *employee* | ReleasedByID | int |
| | IDVerifiedFlag | boolean |
| | SignatureFile | string |

### JOB_STATUS_HISTORY
D3 Repair Jobs

| | | |
|---|---|---|
| PK | **StatusHistoryID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *job_status* | StatusID | int |
| FK *employee* | ChangedByID | int |
| | ChangedAt | datetime |
| | ChangeNote | string |

### NOTIFICATION
D6 Notifications

| | | |
|---|---|---|
| PK | **NotificationID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *employee* | SentByID | int |
| | RecipientEmail | string |
| | DeliveryStatus | string |
| | SentAt | datetime |
| | FailureReason | string |

### QUOTE
D4 Quotes

| | | |
|---|---|---|
| PK | **QuoteID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *employee* | PreparedByID | int |
| | LaborAmount | decimal |
| | PartsAmount | decimal |
| | TotalAmount | decimal |
| | CustomerDecision | string |
| | DecisionAt | datetime |
| FK *employee* | DecisionByID | int |
| | CreatedAt | datetime |

### PAYMENT
D5 Payments

| | | |
|---|---|---|
| PK | **PaymentID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *employee* | ProcessedByID | int |
| | Amount | decimal |
| | PaymentMethod | string |
| | PaymentStatus | string |
| | PaidAt | datetime |

### JOB_NOTE
D3 Repair Jobs

| | | |
|---|---|---|
| PK | **JobNoteID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *employee* | EmployeeID | int |
| | NoteType | string |
| | NoteText | string |
| | CreatedAt | datetime |

### JOB_PART
D3 Repair Jobs

| | | |
|---|---|---|
| PK | **JobPartID** | int |
| FK *repair_job* | RepairJobID | int |
| FK *part* | PartID | int |
| FK *store* | StoreID | int |
| | Quantity | int |
| | PartStatus | string |
| FK *employee* | RecordedByID | int |
| | RecordedAt | datetime |

### PART
D7 Parts Inventory

| | | |
|---|---|---|
| PK | **PartID** | int |
| FK *supplier* | DefaultSupplierID | int |
| | PartName | string |
| | DeviceModel | string |
| | Grade | string |
| | StandardCost | decimal |
| | ActiveFlag | boolean |

### STORE_PART_STOCK
D7 Parts Inventory

| | | |
|---|---|---|
| PK,FK *store* | **StoreID** | int |
| PK,FK *part* | **PartID** | int |
| | QuantityOnHand | int |
| | ReorderPoint | int |

### INVENTORY_TXN
D7 Parts Inventory

| | | |
|---|---|---|
| PK | **InventoryTxnID** | int |
| FK *stock* | StoreID | int |
| FK *stock* | PartID | int |
| FK *repair_job* | RepairJobID | int |
| FK *purch._order* | PurchaseOrderID | int |
| FK *employee* | RecordedByID | int |
| | TxnType | string |
| | QuantityChange | int |
| | Reason | string |
| | TxnAt | datetime |

### SUPPLIER
D9 Suppliers

| | | |
|---|---|---|
| PK | **SupplierID** | int |
| UK | SupplierName | string |
| | ContactName | string |
| | Phone | string |
| | Email | string |
| | ActiveFlag | boolean |

### PURCHASE_ORDER
D8 Purchase Orders

| | | |
|---|---|---|
| PK | **PurchaseOrderID** | int |
| FK *store* | StoreID | int |
| FK *supplier* | SupplierID | int |
| FK *employee* | CreatedByID | int |
| FK *employee* | ApprovedByID | int |
| IX | POStatus | string |
| | CreatedAt | datetime |
| | SubmittedAt | datetime |
| | ApprovedAt | datetime |
| | PlacedAt | datetime |
| | RejectionReason | string |
| | OrderTotal | decimal |

### PO_LINE
D8 Purchase Orders

| | | |
|---|---|---|
| PK | **POLineID** | int |
| FK *purch._order* | PurchaseOrderID | int |
| FK *part* | PartID | int |
| | QuantityOrdered | int |
| | QuantityReceived | int |
| | UnitCost | decimal |

### PO_JOB_LINK
D8 Purchase Orders

| | | |
|---|---|---|
| PK,FK *purch._order* | **PurchaseOrderID** | int |
| PK,FK *repair_job* | **RepairJobID** | int |

## Notation

exactly one → zero or many

exactly one → one or many

zero or one → zero or one

| | |
|---|---|
| **PK** | primary key (shaded row) |
| **FK table** | foreign key and the table it references |
| **UK / IX** | unique constraint / non-unique index |

FKs to store and employee are listed in each key column; their lines are left off so the diagram stays readable (see the relationship list in the data dictionary).

## DFD data store holding each entity

| | | |
|---|---|---|
| D1 | **Customers & Devices:** | CUSTOMER, DEVICE |
| D2 | **Appointments:** | APPOINTMENT |
| D3 | **Repair Jobs:** | JOB_STATUS, REPAIR_JOB, JOB_STATUS_HISTORY, JOB_NOTE, JOB_PART |
| D4 | **Quotes:** | QUOTE |
| D5 | **Payments:** | PAYMENT |
| D6 | **Notifications:** | NOTIFICATION |
| D7 | **Parts Inventory:** | PART, STORE_PART_STOCK, INVENTORY_TXN |
| D8 | **Purchase Orders:** | PURCHASE_ORDER, PO_LINE, PO_JOB_LINK |
| D9 | **Suppliers:** | SUPPLIER |
| D10 | **Staff & Stores:** | ROLE, STORE, EMPLOYEE |

## Changes from the Sprint 2 ERD

1. EMAIL_TEMPLATE removed: the catalog memo judged editable email templates spurious (no client source); the completion email uses one fixed format, so NOTIFICATION no longer references a template.
2. IntakeCondition moved from DEVICE to REPAIR_JOB: a device's condition is recorded at every visit.
3. STORE gains EmailAddress, the store account the completion email is sent from (client-confirmed #7).
4. PURCHASE_ORDER gains SubmittedAt, PlacedAt and RejectionReason (FR10 reject-to-Draft; UC-10 Ordered).
5. APPOINTMENT stays optional for a job (0..1): walk-ins need none (FR01 scope decision, UC-12).
6. Every entity now sits in exactly one DFD data store (store key, left), so the DFDs and the ERD tell the same story.